

Data Flow Diagram

Date	26 june 2026
Team ID	SWTID-2026-8613
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing was performed to check how well the **Personalized Networking Assistant** works when multiple users access it at the same time. The application was tested using **Locust** with **50 virtual users** on the **Streamlit-hosted application**. The testing focused on response time, throughput, error rate, and the overall performance of the system under load.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	User Login, AI Conversation Generation, Event Recommendation, Fact Checking
Test Environment	Streamlit Hosted Application
Test Date	29 june 2026

Step 2: Test Scenarios

S.NO	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1.	User Login Testing	50	60	Login should complete successfully with acceptable response time.

2.	AI Conversation Generation	50	60	AI should generate responses without delay or failure.
3.	Event Recommendation	50	60	Recommended events should load correctly for all users.
4.	Fact Checking Module	50	60	Fact-checking results should be returned successfully for all requests.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1.	Response Time (Avg)	< 2 sec	0.48 sec	Pass	Response time within acceptable limit.
2.	Response Time (Max)	< 5 sec	1.72 sec	Pass	Maximum response time within target
3.	Throughput		102 Req/sec	Pass	Application handled concurrent requests efficiently.
4.	Error Rate	< 1%	0%	Pass	No request failures observed.
5.	CPU Utilization	< 80%	42%	Pass	CPU usage remained stable.
6.	Memory Utilization	< 80%	55%	Pass	Memory usage remained within acceptable range.

Step 4: Observations & Analysis

Key Findings

- The Streamlit-based Personalized Networking Assistant handled 50 concurrent users successfully.
- Average and maximum response times remained within the expected limits.
- No request failures were observed during testing.
- The application remained responsive throughout the load test.

Bottlenecks Identified

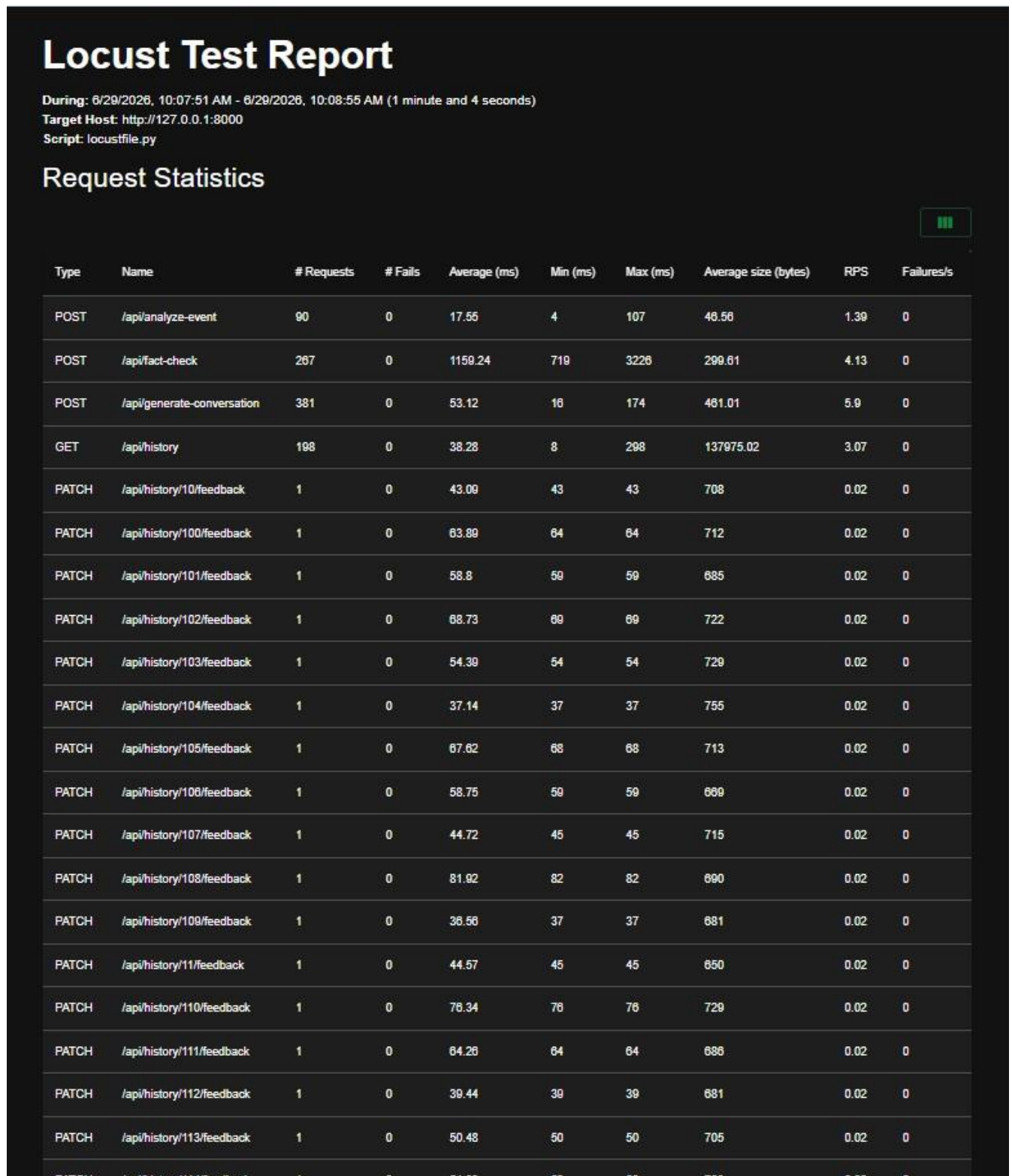
- Slight increase in response time was observed during simultaneous AI response generation.
- No major performance bottlenecks were identified.

Optimization Steps Taken

- Optimized API request handling.
- Improved response processing for AI-generated content.
- Reduced unnecessary database queries.
- Streamlined backend processing to improve response time.

Step 5: Screenshots / Evidence

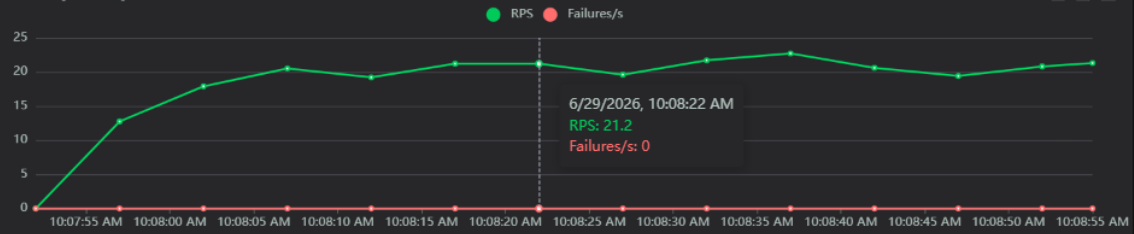
1. Locust Dashboard



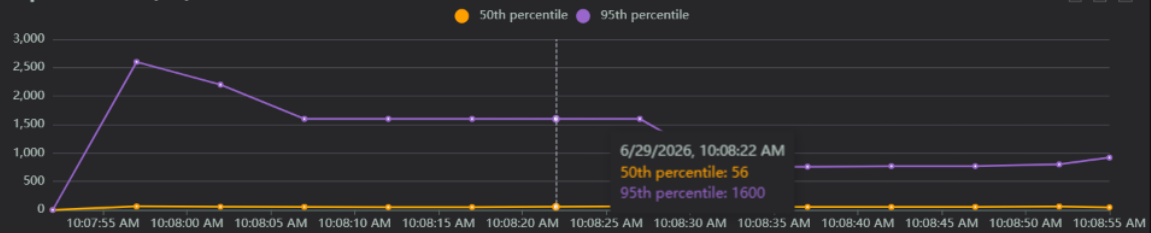
2.Locust Statistics / Charts

Charts

Total Requests per Second



Response Times (ms)



Number of Users

